

ARYA B V

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in aryabv12

Objective

To obtain a challenging position in a reputed organization where I can utilize my technical and problem-solving skills in Electronics and Communication Engineering, while contributing to the company's growth and gaining practical industry experience.

Education

- RV College of Engineering, Bengaluru** 2026
Bachelor of Engineering (B.E), Electronics and Communication Engineering
6.97
- Siddaganaga Polytechnic College, Tumkur** 2023
Diploma in Electrical and Electronics Engineering
9.74
- St. Moses English School, Tumkur** 2020
Class X
79.36

Skills

- Programming Languages: C,C++
- Hardware Description Languages (HDLs): Verilog, System Verilog
- Software & Simulation Tools: MATLAB, Simulink, Vivado, Cadence Virtuoso, Keil µVision, Arduino IDE
- Core ECE Concepts: Digital Logic Design, Analog Electronics, Signals and Systems, Network Theory, Communication Systems, Microprocessors & Microcontrollers, VLSI Design, Control Systems
- Operating Systems: Windows, Linux (Ubuntu)

Projects

- Energy enerlytics: A grid interactive solar pv system with weather forecasting analysis**
 - Developed an IoT-based smart solar energy management system using ESP32 for real-time monitoring of solar generation, battery SOC, load consumption, and grid interaction.
 - Implemented weather-aware energy optimization that dynamically controls battery charging, energy storage, and grid export decisions using forecast data.
 - Designed a safe and resilient microgrid architecture featuring anti-islanding protection, relay interlock logic, fail-safe operation, and cloud-based monitoring through a web dashboard.
- Automated Bio-Sorting System using Real-Time Object Detection and Robotic Manipulation**
 - Developed an AI-powered biomedical waste segregation system using YOLO object detection, trained on a custom dataset of 45,000+ images, achieving over 90% detection accuracy.
 - Integrated NVIDIA Jetson Nano, ESP32, ultrasonic sensors, and a 5-DOF robotic arm with suction gripper to enable real-time object detection, autonomous pick-and-place sorting, and smart bin monitoring.
 - Designed and deployed an end-to-end embedded AI solution using Python, PyTorch, OpenCV, and Streamlit, providing real-time inference, system control, monitoring, and performance visualization.
- Teach Safe Headgear**
 - Developed an IoT-based smart safety helmet using ESP32, integrating gas, temperature, humidity, pressure, and motion sensors for real-time monitoring of mine worker safety
 - Implemented Wi-Fi based communication to transmit worker location and environmental data to a central

monitoring station, enabling remote tracking and emergency alerts

- Designed a real-time monitoring system with web dashboard visualization and buzzer-based warning mechanisms to detect hazardous gas leakage and unsafe environmental conditions.p

- **BengaluruClean: A Smart waste management ecosystem**

- Developed BengaluruClean, a smart waste management ecosystem using React Native, Firebase, and Google Maps, enabling on-demand waste pickup scheduling, real-time location tracking, and role-based access for citizens, workers, and administrator
- Implemented GPS-based location services and a camera-based Proof of Work system to verify waste collection and improve operational transparency
- Built a real-time cloud backend with waste tracking, admin analytics, and a Green Points reward system to encourage waste segregation.

- **Vaidya Mithra: Ai powered healthcare assistant**

- Developed a full-stack healthcare management platform using React, Firebase Authentication, Firestore, and Gemini AI, supporting patient, doctor, attender, and admin workflows.
- Integrated AI-powered symptom triage and conversational healthcare assistance using the Gemini API to provide non-diagnostic health guidance, risk assessment, and personalized recommendations.
- Implemented real-time appointment scheduling, queue management, notifications, audit logging, and role-based access control, enabling seamless digital healthcare operations

Personal Details

- Date of Birth : 12/02/2005